

Multiple Choice (10 marks)

1. Which of the following is the symplastic pathway for the movement of sucrose from the site of photosynthesis in mesophyll cells into the phloem?
 - (a) Bundle sheath, companion cell, phloem parenchyma, fibers
 - (b) Phloem parenchyma, bundle sheath, fibers, companion cell
 - (c) Bundle sheath, phloem parenchyma, companion cell, sieve tube
 - (d) Bundle sheath, fibers, companion cell, sieve tube
 - (e) Fibers, bundle sheath, companion cell, sieve tube
2. What prevents the stomach from being digested by its own secretions?
 - (a) The thick muscular wall of the stomach
 - (b) Antibacterial properties of stomach acid
 - (c) Gastric juice
 - (d) High pH levels in the stomach
 - (e) Presence of beneficial bacteria
3. What is the physiological purpose of shivering?
 - (a) Shivering is a method the body uses to relieve pain.
 - (b) Shivering is a tactic to ward off predators by appearing agitated.
 - (c) Shivering is a way to burn excess calories
 - (d) Shivering is a response to fear or anxiety
 - (e) Shivering is an involuntary action to prepare muscles for physical activity.
4. What are some of the cues used by animal in migrating?
 - (a) Taste, wind direction, humidity
 - (b) Atmospheric oxygen levels, plant growth, tidal patterns
 - (c) Infrared radiation, sound frequency, barometric pressure
 - (d) Moon phases, echo location, gravitational pull
 - (e) Sunlight intensity, water current, soil composition
5. In certain abnormal conditions, the stomach does not secrete hydrochloric acid. What effects might this have on the digestive process?
 - (a) Without HCl, more bacteria would be killed in the stomach.
 - (b) No HCl would lead to reduced mucus production in the stomach lining.

- (c) The absence of HCl would cause the stomach to digest its own tissues.
- (d) No HCl in the stomach would significantly increase the pH of the small intestine.
- (e) The absence of HCl would speed up the digestive process.

True / False (10 marks)

Answer True or False

- 6. True or False: Does anterior laxity of the uninjured knee influence clinical outcomes of ACL reconstruction?
- 7. True or False: Are UK radiologists satisfied with the training and support received in suspected child abuse?
- 8. True or False: Must early postoperative oral intake be limited to laparoscopy?
- 9. True or False: Can the prognosis of polymyalgia rheumatica be predicted at disease onset?
- 10. True or False: Thrombosis prophylaxis in hospitalised medical patients: does prophylaxis in all patients make sense?

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

- 11. Discuss the challenges of using human subjects in inheritance studies, focusing on genetic complexity, environmental factors, ethical considerations, and demographic limitations.
- 12. Discuss the factors that promote linkage disequilibrium in populations and analyze why random mating does not contribute to this phenomenon.

End of Paper